

# Analysis and Validation Report

Service Department Operational Optimization Simulator

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Supporting data: documents/m4/ and results/m4/ (n=30 replications per case, seed policy seed+i).

## 1. Introduction

This report documents Milestone 4 analysis for a discrete-event simulation (DES) of an automotive dealership service department. The simulator, completed in M3, models non-stationary customer arrivals (dealership-day Cox/Poisson process), service-advisor intake, technician/bay repair with Gamma service times and experience scaling, and a parts department with reorder lead times. Per-run metrics and CSV/JSON exports already existed; M4 adds multi-replication statistics, one-at-a-time sensitivity sweeps, multi-factor scenario tests, and structured validation evidence.

**Analysis goals.** (1) Quantify which controllable parameters most drive customer wait and shop utilization. (2) Compare realistic operating scenarios (normal, peak, understaffed, parts stress, ideal capacity). (3) Validate directional reasonableness via face checks, extreme conditions, parameter justification, and soft M/M/c output comparison. (4) Report means, standard deviations, and 95% confidence intervals for primary metrics.

**Baseline configuration** (M3 defaults):  $\lambda = 4$  vehicles/h,  $c = 3$  technicians/bays,  $\mu = 2$  jobs/tech/h, advisors = 4, Gamma shape  $k = 4$ , experience  $\alpha = 1$ , horizon = 10 h, parts lead time  $L = 2$  h, base seed = 42. Replication  $i$  uses seed  $42+i$ . Statistics use  $CI = \text{mean} \pm 1.96 \cdot s/\sqrt{n}$  (Appendix A.3).

## 2. Parameter Analysis

### 2.1 Method

Sensitivity analysis varies one input at a time while holding other baseline parameters fixed. Each level runs  $n = 30$  replications. Sensitivity follows Appendix A.2: Sensitivity =  $(\% \Delta \text{ output}) / (\% \Delta \text{ input})$  relative to the baseline level. Primary outputs are mean total customer wait  $W_{\text{total}}$  (h), shop technician utilization, and jobs completed.

Swept parameters: arrival rate  $\lambda \in \{2, 4, 6, 8\}$ ; technicians  $c \in \{2, 3, 4, 5\}$ ; service rate  $\mu \in \{1.5, 2.0, 2.5\}$ ; advisors  $\in \{1, 2, 4\}$ ; parts lead time  $L \in \{0.5, 2.0, 4.0\}$  hours.

### 2.2 Results

| Parameter       | Level | Wait mean (h) | Wait sens. | Util mean | Util sens. | Jobs mean |
|-----------------|-------|---------------|------------|-----------|------------|-----------|
| arrivalRate     | 2.0   | 0.341         | 0.312      | 0.212     | 0.989      | 19.1      |
| arrivalRate     | 6.0   | 0.501         | 0.482      | 0.621     | 0.956      | 57.1      |
| arrivalRate     | 8.0   | 0.693         | 0.714      | 0.788     | 0.875      | 73.7      |
| technicianCount | 2.0   | 0.543         | -1.030     | 0.628     | -1.485     | 38.1      |
| technicianCount | 4.0   | 0.362         | -0.313     | 0.324     | -0.687     | 39.4      |
| technicianCount | 5.0   | 0.356         | -0.177     | 0.264     | -0.559     | 38.8      |

| Parameter          | Level | Wait mean (h) | Wait sens. | Util mean | Util sens. | Jobs mean |
|--------------------|-------|---------------|------------|-----------|------------|-----------|
| serviceRate        | 1.5   | 0.567         | -1.613     | 0.550     | -1.238     | 37.9      |
| serviceRate        | 2.5   | 0.319         | -0.843     | 0.338     | -0.782     | 37.9      |
| advisorCount       | 1.0   | 0.415         | -0.035     | 0.416     | 0.014      | 37.9      |
| advisorCount       | 2.0   | 0.397         | 0.034      | 0.417     | 0.015      | 37.9      |
| partsLeadTimeHours | 0.5   | 0.384         | 0.067      | 0.420     | 0.001      | 37.9      |
| partsLeadTimeHours | 4.0   | 0.465         | 0.152      | 0.422     | 0.004      | 37.9      |

Table 1. One-at-a-time sensitivity coefficients and mean outputs (n=30). Baseline reference levels:  $\lambda=4$ ,  $c=3$ ,  $\mu=2$ , advisors=4,  $L=2$  h.

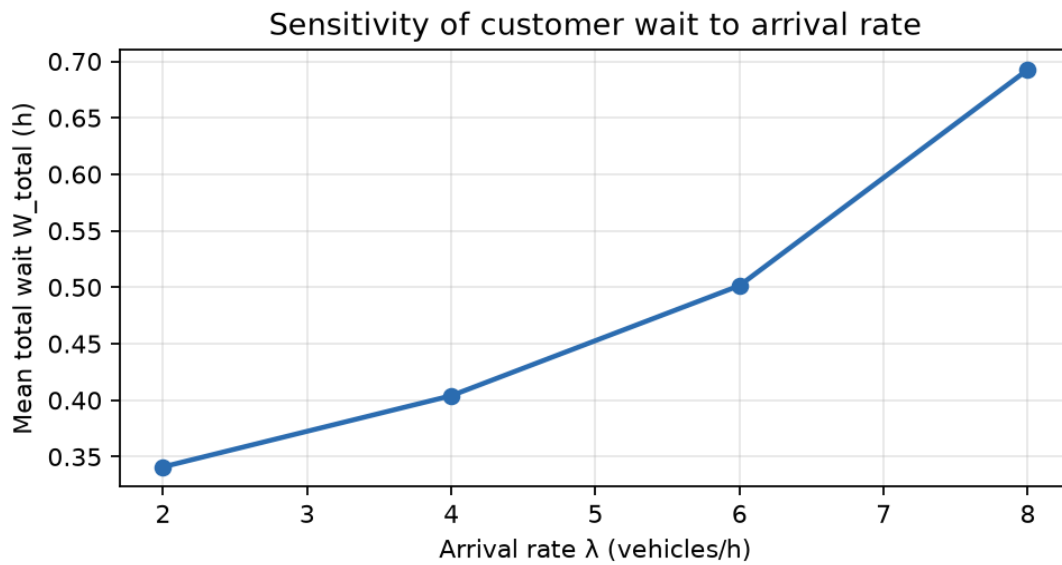


Figure 1. Mean total customer wait versus arrival rate  $\lambda$  (baseline wait at  $\lambda=4$  included).

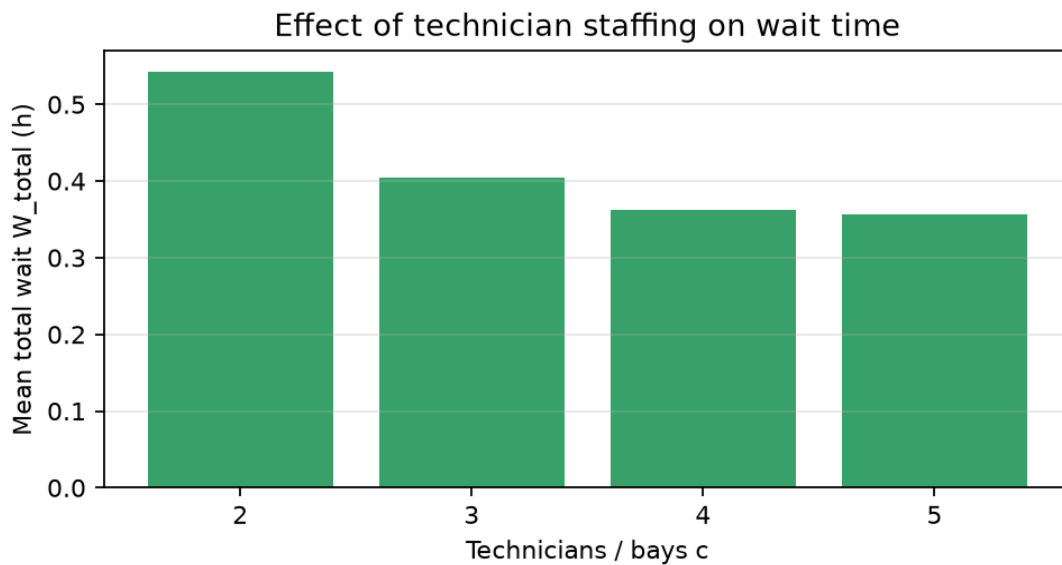


Figure 2. Mean total customer wait versus technician/bay count  $c$ .

## 2.3 Interpretation and ranking

Ranking drivers of wait time (by magnitude of wait sensitivity near the tested steps):

1. **Service rate  $\mu$**  — reducing  $\mu$  from 2.0 to 1.5 raises wait to 0.567 h (wait sensitivity  $\approx -1.61$ ). Slower mean service is the strongest lever on  $W_{\text{total}}$ .
2. **Technician count  $c$**  — cutting staffing from 3 to 2 raises wait to 0.543 h (sensitivity  $\approx -1.03$ ) and utilization (sensitivity  $\approx -1.48$ ). Adding a fourth tech helps, but returns diminish by  $c=5$  (wait 0.356 h vs 0.362 h at  $c=4$ ).
3. **Arrival rate  $\lambda$**  — wait and utilization rise roughly with demand (at  $\lambda=8$ , wait 0.693 h, util 0.788; jobs  $\approx 73.7$ ). Throughput sensitivity  $\approx 1$ , as expected for an unsaturated shop.
4. **Parts lead time  $L$**  — moderate effect on wait (0.384 h at  $L=0.5$  vs 0.465 h at  $L=4$ ); larger effect appears when stock/reorder are stressed jointly (Section 3).
5. **Advisor count** — at baseline  $\lambda$ , reducing advisors to 1 barely changes total wait (0.415 h) but clearly increases advisor wait (0.045 h). Advisors become binding mainly when intake capacity is reduced together with tech capacity (understaffed scenario).

### 3. Scenario Testing

#### 3.1 Scenario definitions

Scenarios change multiple parameters simultaneously to represent realistic shop conditions aligned with M1 expected outcomes (morning rush, staffing shortfall, parts blocking).

- **Normal day (baseline):** defaults above.
- **Peak / rush demand:**  $\lambda = 8$  with baseline staff  $\rightarrow$  long queues, high  $\rho$ .
- **Understaffed shop:**  $c = 2$  technicians and 1 advisor at  $\lambda = 4 \rightarrow$  dual bottleneck.
- **Parts stress:** longer lead time with tighter inventory  $\rightarrow$  elevated  $D_{\text{parts}}$  and “phantom” under-utilization relative to wait.
- **Ideal capacity:**  $c = 5$  technicians (extra capacity) at baseline demand  $\rightarrow$  shorter waits, diminishing returns.

#### 3.2 Results

| Scenario       | W_total | 95% CI wait    | Tech util | D_parts (h) | Jobs/day | Adv wait | D_queue |
|----------------|---------|----------------|-----------|-------------|----------|----------|---------|
| Normal         | 0.404   | [0.367, 0.441] | 0.420     | 0.022       | 37.9     | 0.0001   | 0.050   |
| Peak           | 0.693   | [0.590, 0.795] | 0.788     | 0.047       | 73.7     | 0.0004   | 0.324   |
| Understaffed   | 0.545   | [0.465, 0.625] | 0.616     | 0.023       | 38.1     | 0.0391   | 0.160   |
| Parts stress   | 1.271   | [1.064, 1.477] | 0.428     | 0.870       | 37.9     | 0.0001   | 0.063   |
| Ideal capacity | 0.356   | [0.339, 0.373] | 0.264     | 0.016       | 38.8     | 0.0000   | 0.001   |

Table 2. Scenario comparison means ( $n=30$ ). Wait CI uses 95% normal approximation. Units: hours except jobs (count over 10 h horizon, labeled jobs/day for planning).

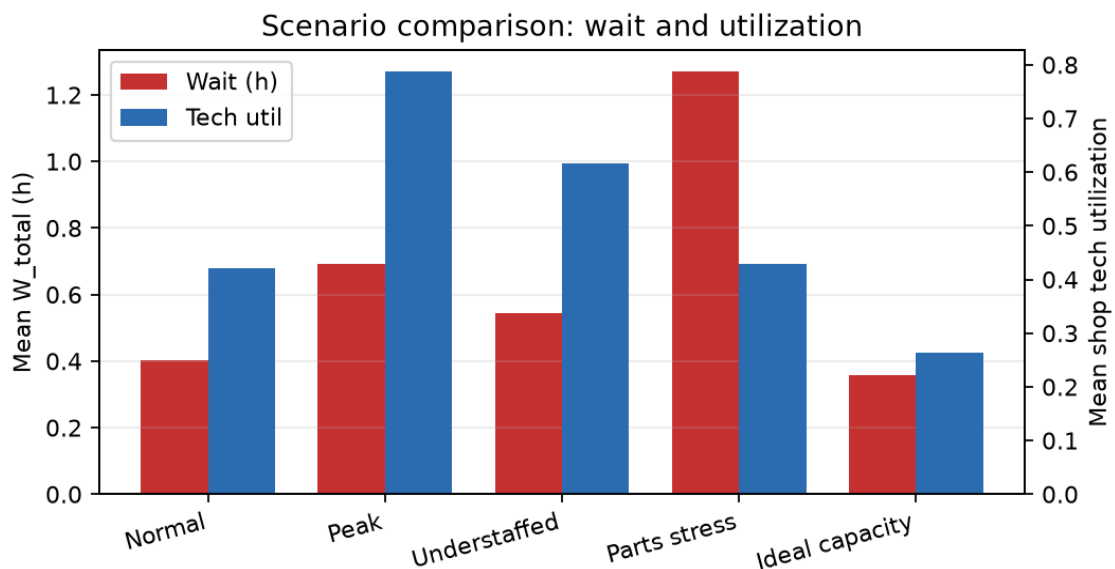


Figure 3. Side-by-side scenario means for total wait and technician utilization.

#### 3.3 Comparison insights

Peak demand nearly doubles wait versus normal (0.693 h vs 0.404 h) and raises shop utilization from 0.420 to 0.788; queue delay jumps to 0.324 h. Jobs completed roughly double (73.7 vs 37.9), confirming the shop absorbs more volume under heavy congestion.

Understaffing increases wait to 0.545 h. Advisor wait rises from near-zero to 0.039 h and queue delay to 0.160 h, showing a compound bottleneck: customers wait both at intake and for technicians. Jobs stay near baseline (~38), so lost service quality appears as delay, not lost completions within the horizon.

Parts stress produces the largest wait (1.271 h) with parts delay 0.870 h, while technician utilization remains near baseline (0.428). This matches the M1 “phantom under-utilization” story: staff appear available while jobs are blocked on parts.

Ideal capacity trims wait to 0.356 h and utilization to 0.264 with negligible queue delay. Marginal gain versus four technicians is small, suggesting advisor/parts constraints and arrival variability bound further improvement under baseline demand.

## 4. Validation

Four validation approaches were applied. Evidence is summarized from documents/m4/validation\_evidence.md and validation\_checks.csv.

### 4.1 Face validation

Expected directional behaviors from domain knowledge were confirmed: (i) peak wait (0.693 h) > baseline (0.404 h); (ii) understaffed wait (0.545 h) > baseline; (iii) ideal-capacity wait (0.356 h) < baseline; (iv) parts-stress D\_parts (0.870 h) ■ baseline (0.022 h). Heuristic checks in Table 3 all pass.

| Check                              | Value | Pass | Notes                   |
|------------------------------------|-------|------|-------------------------|
| face_peak_wait_gt_baseline         | 0.693 | true | peak vs baseline        |
| face_ideal_wait_lt_baseline        | 0.356 | true | ideal vs baseline       |
| extreme_low_arrival_util_near_zero | 0.024 | true | expect util << baseline |
| extreme_overload_wait_grows        | 5.998 | true | expect large waits      |

Table 3. Face/extreme validation heuristic checks.

### 4.2 Extreme conditions

- **Near-empty shop ( $\lambda \approx 0.2$ ):** mean tech utilization 0.024; jobs  $\approx 1.0$  — idle shop.
- **Overload ( $\lambda \blacksquare c_{\mu}$ ):** mean wait 5.998 h, jobs  $\approx 195$  — queues explode as expected.
- **Single technician ( $c=1$ ):** wait 2.613 h, util 1.323 — severe congestion.
- **Instant parts ( $L \approx 0$ ):** D\_parts mean 0.000 h — parts module behaves correctly.

### 4.3 Parameter validation

Defaults match the M3 parameter table and M1 small-dealership illustration: a few technicians, Poisson-style arrivals with a dealership-day profile, Gamma service (shape 4) for moderate variability, and experience scaling. Values are experience-based / illustrative rather than fitted to a proprietary dealership telemetry set (no real site dataset was available).

### 4.4 Output comparison (M/M/c soft benchmark)

Analytical M/M/c  $\rho$  for the baseline is 0.667, while simulated shop utilization averages 0.420 (relative error mean 0.370, 95% CI [0.322, 0.417]). Zero of 30 baseline replications meet a strict overallValid flag under the M3 tolerance. This mismatch is *expected* and informative: the live model includes an advisor stage, Gamma (not exponential) service, experience factors, non-stationary arrivals, and parts blocking—none of which the classic M/M/c formulas include. M/M/c is therefore used as a soft structural benchmark, not a pass/fail oracle.

## 4.5 Limitations

- No warm-up deletion; metrics include the full 10 h horizon.
- No empirical dealership arrival/service dataset for calibration.
- FIFO queues only; bay count tied to technicians.
- Constant reorder policy; no dynamic expediting or multi-SKU complexity.
- Analytical M/M/c comparison is intentionally honest about structural mismatch.

## 5. Statistical Summary

Primary metrics (matching the M3 tracked set): average total customer wait  $W_{total}$ , shop technician utilization, parts delay  $D_{parts}$ , throughput (jobs completed), and advisor wait / queue delay as bottleneck indicators. Table 4 reports the required statistical columns for the baseline case; Tables 5–6 summarize selected scenario metrics.

### 5.1 Baseline (normal day)

| Metric           | n  | Mean   | Std Dev | Min    | Max    | 95% CI           |
|------------------|----|--------|---------|--------|--------|------------------|
| $W_{total}$ (h)  | 30 | 0.404  | 0.103   | 0.253  | 0.742  | [0.367, 0.441]   |
| Tech util        | 30 | 0.420  | 0.089   | 0.243  | 0.618  | [0.388, 0.452]   |
| $D_{parts}$ (h)  | 30 | 0.022  | 0.035   | 0.000  | 0.129  | [0.010, 0.035]   |
| Jobs completed   | 30 | 37.933 | 5.795   | 24.000 | 49.000 | [35.860, 40.007] |
| Advisor wait (h) | 30 | 0.000  | 0.000   | 0.000  | 0.002  | [-0.000, 0.000]  |
| $D_{queue}$ (h)  | 30 | 0.050  | 0.068   | 0.001  | 0.310  | [0.025, 0.074]   |

Table 4. Baseline statistical summary (n=30, seed 42...71).

### 5.2 Peak and parts-stress scenarios

| Metric           | n  | Mean   | Std Dev | Min    | Max    | 95% CI           |
|------------------|----|--------|---------|--------|--------|------------------|
| $W_{total}$ (h)  | 30 | 0.693  | 0.287   | 0.324  | 1.301  | [0.590, 0.795]   |
| Tech util        | 30 | 0.788  | 0.122   | 0.567  | 1.010  | [0.744, 0.831]   |
| $D_{parts}$ (h)  | 30 | 0.047  | 0.075   | 0.000  | 0.351  | [0.020, 0.074]   |
| Jobs completed   | 30 | 73.700 | 7.140   | 59.000 | 88.000 | [71.145, 76.255] |
| Advisor wait (h) | 30 | 0.000  | 0.001   | 0.000  | 0.003  | [0.000, 0.001]   |
| $D_{queue}$ (h)  | 30 | 0.324  | 0.249   | 0.049  | 0.936  | [0.235, 0.413]   |

Table 5. Peak / rush scenario statistics (n=30).

| Metric           | n  | Mean   | Std Dev | Min    | Max    | 95% CI           |
|------------------|----|--------|---------|--------|--------|------------------|
| $W_{total}$ (h)  | 30 | 1.271  | 0.577   | 0.336  | 2.934  | [1.064, 1.477]   |
| Tech util        | 30 | 0.428  | 0.091   | 0.236  | 0.635  | [0.395, 0.461]   |
| $D_{parts}$ (h)  | 30 | 0.870  | 0.564   | 0.011  | 2.515  | [0.668, 1.071]   |
| Jobs completed   | 30 | 37.933 | 5.795   | 24.000 | 49.000 | [35.860, 40.007] |
| Advisor wait (h) | 30 | 0.000  | 0.000   | 0.000  | 0.002  | [-0.000, 0.000]  |
| $D_{queue}$ (h)  | 30 | 0.063  | 0.053   | 0.003  | 0.212  | [0.044, 0.082]   |

Table 6. Parts-stress scenario statistics (n=30).

### 5.3 Discussion of variability

Baseline wait SD is 0.103 h against a mean of 0.404 h ( $CV \approx 26\%$ ), so seed variation matters but the 95% CI [0.367, 0.441] remains informative with n=30. Peak and parts-stress cases show higher absolute SD (wait SD

0.287 h and 0.577 h), consistent with near-capacity or inventory blocking amplifying path dependence. Jobs completed under baseline has  $SD \approx 5.8$  around mean 37.9; capacity and arrival volatility both contribute. Negative CI bounds for near-zero advisor wait are an artifact of the symmetric normal approximation and should be interpreted as  $\approx 0$ .

## 6. Conclusions

### 6.1 Main findings

- Service rate and technician staffing dominate wait-time sensitivity near the baseline; arrival rate scales both wait and throughput.
- Parts stress can inflate total wait far more than bare lead-time sensitivity suggests when inventory policy is tight, while utilization stays deceptively moderate.
- Understaffing surfaces advisor delays that are invisible at four advisors.
- Extra technicians past four yield diminishing wait reductions at baseline demand.
- Face and extreme tests support model reasonableness; M/M/c disagreement is explained by structural features beyond multi-server exponential theory.

### 6.2 Strengths and weaknesses

**Strengths:** end-to-end DES with wait decomposition; reproducible batch harness (seed+i);  $n=30$  statistical rigor; scenarios that isolate staffing vs inventory stories; honest soft validation against M/M/c.

**Weaknesses:** illustrative parameters without field telemetry; no warm-up discard; simplified parts/FIFO logic; utilization definitions under extreme overload can exceed 1 when busy-time accounting includes backlog effects—treat overload util as a stress signal rather than a bounded fraction.

### 6.3 Recommendations

For capacity planning with this tool: (1) protect mean service productivity (training / experience) before over-hiring beyond the arrival rate; (2) staff technicians for peak  $\lambda$  scenarios, not only average days; (3) treat parts policy as a first-class lever—short lead time or safer stock can cut wait without raising apparent utilization; (4) when advisor count drops, monitor advisor wait separately from shop queue delay; (5) use replications ( $n \geq 30$ ) before trusting a single UI screenshot run.

```
Reproducibility:      mvn      -q      -DskipTests      compile      exec:java
"-Dexec.mainClass=simulator.analysis.M4ExperimentRunner"
"-Dexec.args=--replications=30      --output=results/m4"
"-Djava.util.logging.config.file=logging-m4.properties". Regenerate this PDF via python
documents/m4/generate_m4_report.py.
```